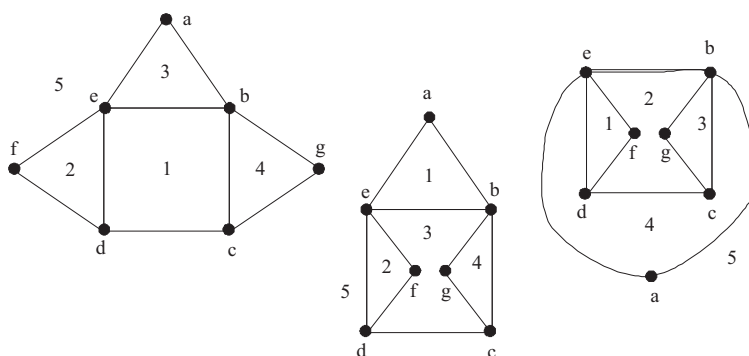


Study theorem 2 (Euler's formula) which is also given below (but skip its proof), and example 5 to see how it is applied.

Theorem 2 (Euler's formula, 1752) *If G is a connected planar graph, then any plane graph depiction of G has $r = e - v + 2$ regions.*

The following three plane depictions of the same graph illustrates Euler's theorem:



Each of the three plane depictions has the same number of regions. (A region of a graph is sometimes also referred to as a face of a graph, and it is just an area of a graph which is surrounded by edges.) The graph above has a total of five regions (four inner regions) and the outer region as indicated in the sketch above. Study the corollary of Euler's formula (with its proof). Note that the hypothesis in this corollary requires that the graph has to be connected, planar, and with at least two edges. We repeat the corollary below:

Corollary *If G is a connected planar graph with $e > 1$, then $e \leq 3v - 6$.*

This Corollary implies that if G is a graph with *more* than $3v - 6$ edges, then G is nonplanar.

This is therefore an easy way of determining whether it is NONplanar.

It is very important to note that it does NOT say that if G is a graph with *at most* $3v - 6$ edges, then G is planar.

We'll show below that the requirement of connectedness is not necessary. However, note that the corollary does not hold for a graph with only one edge, such as K_2 (check this!). Note where in the proof the assumption that the graph has more than one edge is used: Where it is claimed that each region has degree ≥ 3 .

We now present two extensions to this corollary. We still call them corollaries of Euler's formula, since they are also simple consequences of Euler's formula. The first one shows that it is not necessary to assume that the graph is connected.

Corollary. *If G is a planar graph with $e > 1$, then $e \leq 3v - 6$.*

Proof. If G is connected, we may simply apply the first corollary of Euler's formula. Otherwise, let the connected components of G be $G_1, G_2, \dots, G_k$, with $k \geq 2$ the number of connected components of G . Consider a drawing of G in the plane. It is possible to join one of the vertices of G_1 to one of the vertices of G_2 without having the edges cross. Similarly, join G_2 and G_3 with an edge, as well as G_3 and $G_4, \dots, G_{k-1}$ and G_k . Denote the new graph obtained in this way by G' . The number of its vertices is the same as that of G , namely v . The number of its edges is $k - 1$ more than that of G : $e + k - 1$. Since G' is connected and still planar, and it definitely has more than one edge, we may apply the first corollary to it to obtain

$$e + k - 1 \leq 3v - 6.$$

Since $k > 1$, we also have

$$e < e + k - 1,$$

and putting together the two inequalities, we obtain

$$e < 3v - 6$$

in the case where G is disconnected.

Therefore, the inequality $e \leq 3v - 6$ holds for all planar graphs with $e > 1$. □

Corollary. *If G is a triangle-free planar graph (ie it does not contain K_3 as a subgraph), with $e > 1$, then $e \leq 2v - 4$.*

When G is connected, the proof is almost identical to the proof of the corollary in Tucker. The difference is that it now follows from the facts that there is more than one edge, and that there are no triangles, that each region has degree ≥ 4 . Then it follows that $2e \geq 4r$, hence $\frac{1}{2}e \geq r$. Combining this with Euler's formula, we obtain

$$\frac{1}{2}e \geq r = e - v + 2,$$

and solving for e we obtain $e \leq 2v - 4$.

When G is disconnected we complete the proof as before (by joining the different components with edges in a plane drawing).

It follows from this corollary that for planar bipartite graphs we have $e \leq 2v - 4$, since bipartite graphs are triangle-free:

Corollary *If G is a planar bipartite graph with $e > 1$, then $e \leq 2v - 4$.*

2.4 Common mistakes

2.4.1 The converse of the $e \leq 3v - 6$ corollary is false

The corollary to Euler's theorem on planar graphs states that

if a graph with v vertices and e edges is planar, then $e \leq 3v - 6$.

The converse is false: if $e \leq 3v - 6$ for a given graph, then you cannot conclude that the graph is planar. Take as example $K_{3,3}$ —it satisfies the inequality (check this!), but it is not planar, as shown in example 3 of section 1.3 in Tucker. (However, it does not contain any triangles, since a triangle is a circuit of odd length, and it does not satisfy $e \leq 2v - 4$, so from the corollary above for triangle-free planar graphs, we again deduce that $K_{3,3}$ is not planar.)